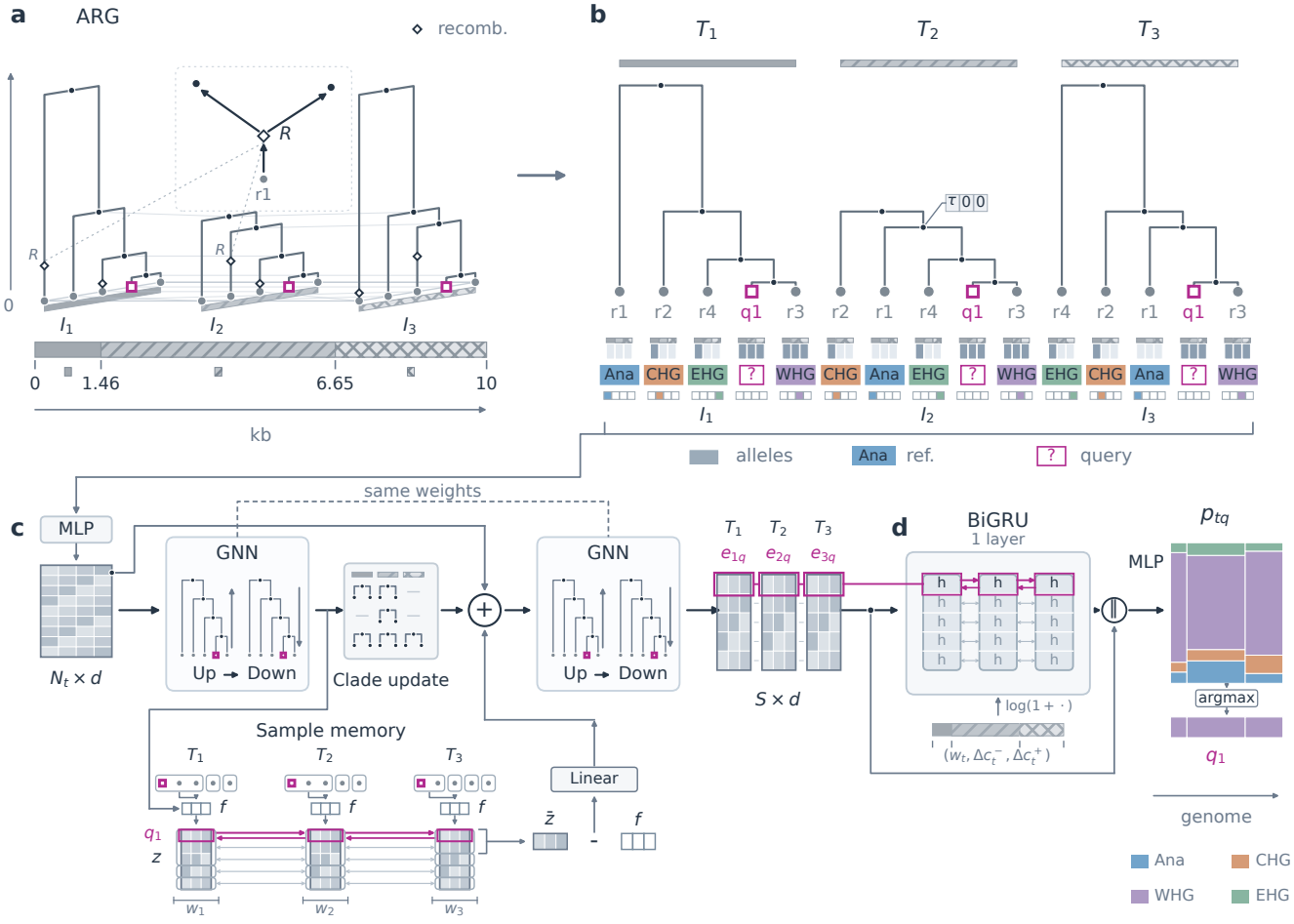

a
ARG
recomb.
time
0
R
r1
I_1
I_2
I_3
0
1.46
6.65
10
kb
b
T_1
T_2
T_3
τ 0 0
r1 r2 r4 q1 r3
Ana CHG EHG ? WHG
alleles
ref.
query
same weights
c
MLP
$N_t \times d$
GNN
Up → Down
Clade update
Sample memory
Linear
q_1
z
w_1
w_2
w_3
$\bar{z}$
d
BiGRU
1 layer
h
log(1 + ·)
$(w_t, \Delta c_t^-, \Delta c_t^+)$
e_{1q} e_{2q} e_{3q}
$S \times d$
MLP
p_{tq}
argmax
q_1
genome
Ana CHG
WHG EHG